

LABORATORY

Lab 2

Distance and Shielding · United States Air Force Academy, Department of Physics and Meteorology

35 Points

NAMES

SCORE

/ 35

The purpose of this lab is to determine the effects of distance and shielding on flux. There are 2 sections to the lab — the first is solely measuring the effects of shielding, and the second is solely measuring the effects of distance. You will be using a ^{137}Cs source, which is similar to what you used in Lab 1. However, unlike Lab 1, your source will include both the initial ^{137}Cs , and the $^{137}\text{Ba}^*$ that it decays into (see decay equations below). You will do most of your analysis in the provided spreadsheet, but only this document will be graded.

I · PART 1: SHIELDING

A. Introduction

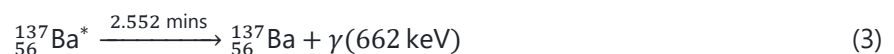
When gamma rays pass through matter, the flux ϕ is given by

$$\phi = \phi_0 e^{-\mu x} \quad (1)$$

after passing through a thickness x of the material, where ϕ_0 is the initial intensity before reaching the material, and μ is the linear attenuation coefficient. See Table A.6 for typical values of mass attenuation coefficients μ/ρ . This experiment will be run similarly to the half-life experiment from Lab 1. The falloff in detector count rate S of gammas as they pass through shields of increasing thickness x will be measured by a Geiger-Mueller detector. From the data, a plot of S vs x will be created and examined to determine the mass attenuation coefficient.

For this part of the lab, we will ignore the inverse square law ($\phi = \frac{S}{4\pi r^2}$). You will do most of your analysis in the provided spreadsheet, but only this document will be graded.

Our ^{137}Cs source has the following decay reactions occurring inside of it. We will be detecting the gamma decays from $^{137}\text{Ba}^*$ in the GM detector.



B. Equipment

- Geiger-Mueller detector
- Timer/counter unit
- Radioactive ^{137}Cs source, obtained from instructor
- Shielding materials, obtained from the instructor.

C. Procedure

1. Connect the cable from the Geiger-Mueller (GM) tube to the back of the counter unit and plug the unit into an outlet. **Do not bring anything into contact with the bottom of the GM tube, it is very fragile.**
2. Set the power supply voltage on the counter to the value on the GM tube (860 V if not labeled).
3. Obtain a 2-minute background count, determine the background-counting rate, and record your measurements in the table below.
4. Obtain a radioactive ^{137}Cs source from your instructor, and place the source on the plastic tray at a distance of 5 cm (the fifth slot down).
5. Choose one of the blocks of lead shielding and place it in the slot directly below the detector. Take a 30 second count and record in the table below. Repeat for 4 different block thicknesses.
6. Calculate the counting rate, subtract the background counting rate, and record the result in the table below.
7. Complete the data collection sections for both Part 1 and Part 2 of the lab before moving on to the analysis for either.

D. Data

3 pts

SHIELD THICKNESS	COUNT TIME [S]	COUNT VALUE [#]	COUNT RATE, R [# / S]
Background			
$x =$			
$x =$			
$x =$			
$x =$			

E. Analysis

1. Fill out Section 1 of the Excel spreadsheet with data from your shielding materials and source. Fill out sections 2 and 3 with the data you took.
2. The spreadsheet will plot the detector count rate S vs ρx (density times thickness).
3. Do the calculations for the bottom of Section 3 of the spreadsheet.

3 pts

F. Discussion Questions

4 pts **Question 1.** Using your data, calculate and report the mass attenuation coefficient μ/ρ for your shielding material. Show the full derivation for how you determined what value you should take off of your plot.

3 pts **Question 2.** Find the accepted value for μ/ρ for your shielding material and report it below. Given the uncertainty in your μ/ρ , is your value consistent with the accepted value you found? Quantifiably justify.

3 pts **Question 3.** How do we know that we are just seeing the gammas from $^{137}\text{Ba}^*$ in the detector, and not the beta particles from ^{137}Cs ?

3 pts

Question 4. Why can we ignore the $1/r^2$ rule and still get a good answer?

3 pts

Question 5. If our detector is not very efficient (e.g. it only counts 50% of the gammas that pass through it), should that affect your conclusion?

II · PART 2: DISTANCE

A. Introduction

Most radioactive sources are isotropic in nature. This means that any radiation is given off equally in all directions. In the case of an isotropic source, it is well known that the intensity of the source falls off as r^2 . This experiment verifies this inverse square relationship for a ^{137}Cs source. Recall the inverse square law:

$$\phi = \frac{S}{4\pi r^2} \quad (4)$$

B. Equipment

- Geiger-Mueller detector
- Timer/counter unit
- Radioactive ^{137}Cs source, obtained from instructor
- 1-meter stick

C. Procedure

1. Lay the GM tube on a flat surface next to a 1-meter stick.
2. If you haven't already done so, obtain a 2-minute background count, determine the background-counting rate, and record your measurements in the table below. If you already took a background count, use your previous data.
3. Obtain a radioactive ^{137}Cs source from your instructor. Choose a distance between 10–100 cm, and place the source at this distance so it is facing the GM detector.
4. Take a 30 second count and record in the table below. Repeat for 4 different distances.
5. Calculate the counting rate, subtract the background counting rate, and record the result in the table below.
6. Complete the data collection sections for both Part 1 and Part 2 of the lab before moving on to the analysis for either.

D. Data**3 pts**

SHIELD THICKNESS	COUNT TIME [S]	COUNT VALUE [#]	COUNT RATE, R [# / S]
Background			
$x =$			
$x =$			
$x =$			
$x =$			

E. Analysis

1. Record the data in Section 4 of the Excel spreadsheet.
2. The spreadsheet will generate a log-log plot of flux vs distance, which are related by the inverse square law.

F. Discussion Questions

3 pts

Question 1. We will be using the data you collected to try to prove that the power of r in the inverse square law is 2 (or close to 2). Starting with the inverse square law, derive the log-log equation that should be represented in the plot.

3 pts

Question 2. Based on your equation above, what is the actual value of the power of r produced from your data? Use a best-fit line to the data to determine this, and write both the full equation and which variable represents the power of r below.

4 pts

Question 3. From your radiation safety training, the three protective measures against external radiation sources are time, distance, and shielding. Discuss the effectiveness of each technique in the context of the information measured in lab 1 and 2. Which are the easiest to control? In what situations might you need to use each?

TURN IN

Send your spreadsheet to your instructor directly, and upload your completed lab write-up to Gradescope (make sure you include all of your lab partners' names when you upload it to Gradescope — digitally link them, don't just include their names on the write-up).